

Resumo de Eletromag

forças

$$F_m = q \vec{V} \times \vec{B}$$

$$F_m = i d\vec{l} \times \vec{B}$$

$$\vec{\tau} = \vec{m} \times \vec{B} \quad \vec{m} = N i \vec{A}$$

curas especiais

hall proposição



$$i = N A V_d e$$

$$i \rightarrow \vec{B}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} i \frac{d\vec{l} \times \hat{r}}{r^2}$$

$$\vec{B} = \frac{\mu_0}{4\pi} q \frac{\vec{V} \times \hat{r}}{r^2}$$

$$\oint B dl = \mu_0 I_{enc} \tau$$

curas especiais

- espira
 - fio ao e $\frac{1}{2} \pi$ circular
 - Solenóide ret
 - Toróide circ
 - Cícl. longo circ
 - placa so ret
- $i = L V \delta$

$$\vec{B} \rightarrow i$$

$$\oint E \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$$

Norte Sai Sul entra

$$W = - \int P_{ext} dv$$

100% rmi $\rightarrow P_{ext} = P_{int}$

$$\rightarrow W = -n R \Delta T$$

$$\bigcirc \rightarrow \bigcirc \rightarrow \cdot$$

PV

$$H = U + PV$$
$$\Delta H = \Delta U + P\Delta V$$

$$Q_p = \Delta H \quad Q_v = \Delta U \quad C_p - C_v = R \quad |a_v| > |a_p|$$

Entropia $S = k \ln W$ $dS = \frac{dQ}{T}$

$p \mid w_{rev} p \mid rev$

Quero 5 Grunhos $\rightarrow$ em seu estresse
de circunstâncias e
não há tempo para tudo
de cada

expanded IsoTernice

$$\Delta \varepsilon_c = 0 \quad \Delta \varepsilon_p = 0 \quad \Delta U = 0$$

4. $\Delta \epsilon_p = 0 \quad \Delta \epsilon_c \neq 0$

função anti difus
 $\Delta \varepsilon_p = 0$ $P_{\text{ext}} = CTE$

Expansive line
post - 0

$$\rightarrow Q_2 = -W$$

$$\Delta S = nR \ln \frac{v_f}{v_o}$$

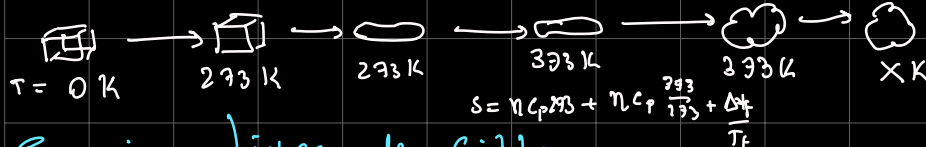
$$\Delta S = n R_{\text{div}} \ln \frac{T_f}{T_i} \quad \frac{\Delta H_f}{T_f} ; \frac{\Delta H_c}{T_c} \quad \Delta S > 0$$

Entropia total do Grupo 1 estrutura cristalina

$$S=0$$

$$S = n C_p 273$$

$$S = \eta \left(p_2 \dot{V}_2 + \frac{\dot{M} h}{T_0} \right)$$



Énergie libre de Gibbs

$$\Delta G = \Delta H - T\Delta S$$

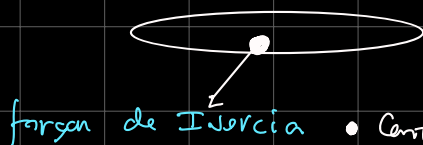
- $\Delta G < 0$ spontaneous
- $\Delta G = 0$ equilibrium
- $\Delta G > 0$ non spontaneous

Dani

Lib é linda
e maravilhosa

Mec Rot

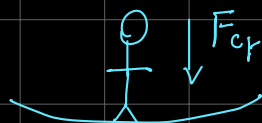
Objetos sujeitos a Rot



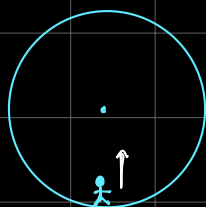
força de Inércia

• Centrífuga $m \frac{v^2}{R} = m \omega^2 R$

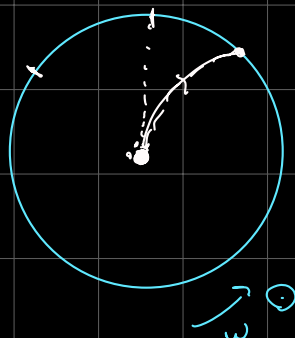
• Coriolis $2m(\vec{v} \times \vec{\omega})$
força de vc
arbitrariamente no disco de per



$$m \omega^2 R = F_{Cp}$$



$$m \omega^2 R + 2m(\vec{v} \times \vec{\omega}) = N$$



análogos

$$v = \omega R$$

Massa $I = \int r^2 dm$

Partícula (anel) casca Cilíndrica

$$I = m R^2$$

Disco/cilindro Maciço

$$I = \frac{m R^2}{2}$$

Barras

esfera $\frac{m r^2}{5}$ ou $\frac{2}{5} m r^2$

$$I_c = I_{cm} + m d^2$$

momento $L = \vec{r} \times \vec{p}$

Energia Cinética $\pm \frac{\omega^2}{2}$

Ondulatória

$$v = \sqrt{\frac{T}{\mu}} \text{ em Corda}$$

$$\Gamma = \frac{A_r}{A_i} \quad \text{Extremidade livre} \quad \text{fixa}$$

$$\Gamma = 1$$

$$\Gamma = -1$$

$$\Gamma = \frac{A_r}{A_i} \quad \text{Corda fixa} \quad \text{fim e Corda}$$

$$\Gamma = \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1} + \sqrt{\mu_2}} \quad \Gamma = \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}} \quad |\Gamma| - \Gamma = 1$$

$$V_y = -\omega A \sin(\omega t - kx + \phi_0)$$

$$P_{\text{avg}} = 2\mu\pi f^2 A^2 v_{\text{prop}}$$

$$\mathcal{E} = 2\mu\pi^2 f^2 A^2 \ell$$

$$\text{batimento} \quad f_{\text{percebida}} = \frac{f_1 + f_2}{2}$$

$$f_{\text{batimento}} = |f_1 - f_2|$$